

12-3 MIXED GAS DIVING EQUIPMENT / SYSTEMS

Mixed gas diving requires a predetermined supply of breathing gases and carbon dioxide absorbent material. Operations must be planned thoroughly to determine usage requirements in order to effectively obtain required supplies in port or at sea prior to the start of the mission. Logistic requirements may include planning for on-site resupply of mixed gases and other supplies and for relief of diving teams.

12-3.1 Gas Mixtures. Four gas mixtures are required to dive the surface-supplied mixed gas tables over their full range:

- Bottom Mixture - The bottom mixture may vary from 90% helium 10% oxygen to 60% helium 40% oxygen depending on the diver's depth. The allowable range of bottom mixtures for each depth is shown in Table 12-4.
- 50% Helium 50% Oxygen - This mixture is used from 90 fsw to 40 fsw during decompression. Oxygen concentration in the mixture may range from 49 to 51 percent.
- 100% Oxygen - Oxygen is used at the 30-fsw and 20-fsw water stops during inwater decompression and at 50, 40 and 30 fsw in the chamber during surface decompression.
- Air - Air is used as an emergency backup gas throughout the dive and to provide air breaks during oxygen breathing.

Instruments used to measure oxygen content in helium-oxygen mixtures shall be accurate to 0.5 percent.

12-3.2 Flyaway Dive System (FADS) III Mixed Gas System (FMGS). The FADS III Mixed Gas System (FMGS) is a portable, self contained, surface supplied diver life support system designed to support mixed gas dive missions to 300 fsw (Figure 12-2 and Figure 12-3). The FMGS consists of five gas rack assemblies:

- One (1) Air Supply Rack Assembly (ASRA)
- One (1) Oxygen Supply Rack Assembly (OSRA)
- Three (3) Helium-Oxygen Supply Rack Assemblies (HOSRA)

Compressed air is provided by a 5000 psi air compressor assembly, which includes an air purification system. The FMGS also includes a mixed gas control console assembly (MGCCA) and two gas booster pump assemblies to charge the OSRA and HOSRA. Set-up and operating procedures for the FMGS are found in the Operating and Maintenance Technical Manual for Fly Away Dive System (FADS) III Mixed Gas System, S9592-B2-OMI-010.



Figure 12-1. FADS III Mixed Gas System (FMGS).

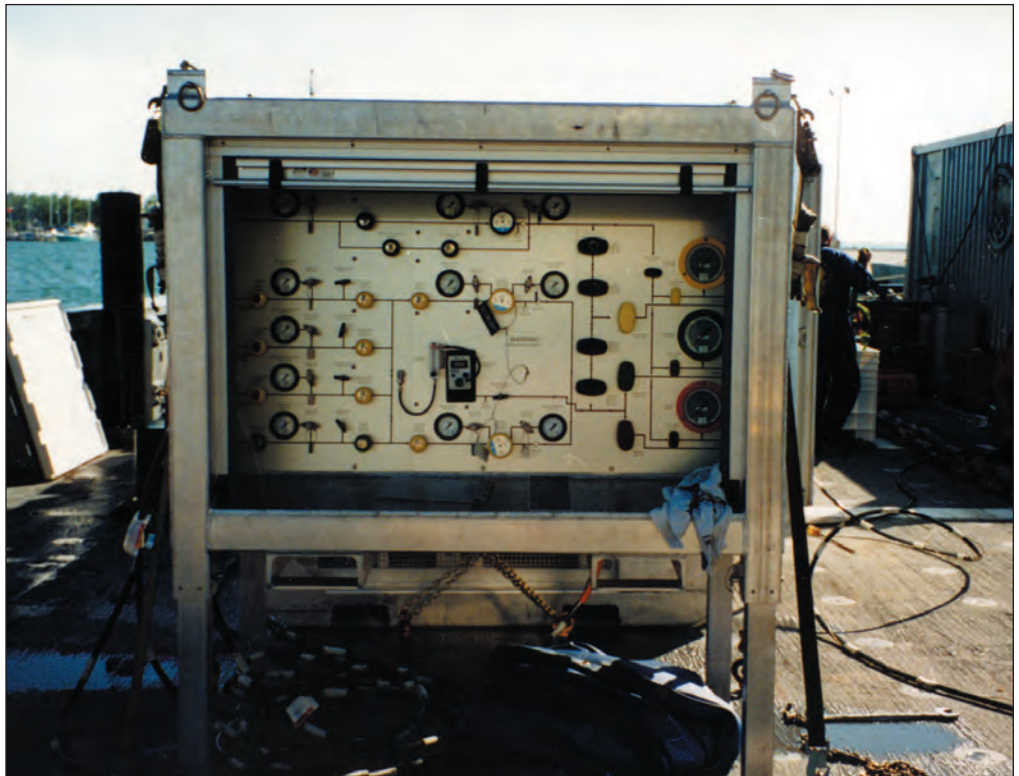


Figure 12-2. FMGS Control Console Assembly.

12-4 SURFACE-SUPPLIED HELIUM-OXYGEN DESCENT AND ASCENT PROCEDURES

The Surface-Supplied Helium-Oxygen Decompression Table ([Table 12-4](#)) is used to decompress divers from surface-supplied helium-oxygen dives. The table is in a depth time format similar to the U.S. Navy Air Decompression Table and is used in a similar fashion.



Figure 12-3. Dive Team Brief for Divers.

- 12-4.1 Selecting the Bottom Mix.** The Surface-Supplied Helium-Oxygen Decompression Table ([Table 12-4](#)) specifies maximum and minimum concentrations of oxygen allowable in the helium-oxygen mixture at depth. The maximum oxygen concentration is set so the diver never exceeds an oxygen partial pressure of 1.3 ata while on the bottom. The minimum oxygen percentage allowed in the mixture is 14 percent for depths to 200 fsw and 10 percent for depths in excess of 200 fsw. Diving with a mixture near maximum oxygen percentage is encouraged as it offers a decompression advantage to the diver. For operational planning, the range of possible depths should be established and a mixture selected that will meet the maximum/minimum specification across the depth range.
- 12-4.2 Selecting the Decompression Schedule.** To select a proper decompression table and schedule, measure the deepest depth reached by the diver and enter the table at the exact or next greater depth. When using a pneumofathometer to measure depth, correct the observed depth reading as shown in [Table 12-2](#). Ensure the pneumofathometer is located at mid-chest level.

Table 12-2. Pneumofathometer Correction Factors.

Pneumofathometer Depth Reading	Correction Factor
0-100 fsw	+1 fsw
101-200 fsw	+2 fsw
201-300 fsw	+4 fsw
301-400 fsw	+7 fsw

Example: The diver's pneumofathometer reads 250 fsw. In the depth range of 201-300 fsw, the pneumofathometer underestimates the diver's depth by 4 fsw. To determine a diver's depth, add 4 fsw to the pneumofathometer reading giving the diver's depth as 254 fsw.

Bottom time is measured as the time from leaving the surface to leaving the bottom, rounded up to the next whole minute, except as noted in [paragraph 12-4.5](#). Enter the table at the exact or next greater bottom time.

- 12-4.3 Travel Rates and Stop Times.** The descent rate is not critical, but in general should not exceed 75 fsw/min. The ascent rate from the bottom to the first decompression stop, between decompression stops, and from the last decompression stop to the surface is 30 fsw/min. Minor variations in the rate of ascent between 20 and 40 fsw/min are acceptable. For surface decompression, the ascent rate from the 40 fsw water stop to the surface is 40 fsw/min.

The time at the first decompression stop begins when the diver arrives at the stop and ends when he leaves the stop. For all subsequent stops, the stop time begins when the diver leaves the previous stop and ends when he leaves the stop. In other words, ascent time between stops is included in the subsequent stop time. The single exception is the first oxygen stop at 30 fsw. The 30-fsw oxygen stop begins when the divers are confirmed to be on oxygen at 30 fsw and ends when the divers leave 30 fsw. The ascent time from the 30- to the 20-fsw oxygen stop is included in the 20-fsw oxygen stop time.

- 12-4.4 Decompression Breathing Gases.** Decompress on bottom mixture to 90 fsw, then shift the diver to a 50% helium 50% oxygen mixture. Upon arrival at the 30 fsw stop, shift the diver to 100% oxygen.

For all dives, surface decompression may be used after completing the 40 fsw water stop as described in [paragraph 12-4.11](#). During surface decompression, the diver surfaces while breathing 50% helium 50% oxygen.

- 12-4.5 Special Procedures for Descent with Less than 16 Percent Oxygen.** To prevent hypoxia, a special descent procedure is required when the bottom mixture contains less than 16% oxygen:

1. Place the diver on air on the surface.

2. Make the appropriate pre-dive checks.
3. Have the diver descend to 20 fsw.
4. At 20 fsw, shift the diver to the bottom mix and ventilate the diver for 20 seconds.
5. Confirm the diver is on bottom mix, then perform a final leak check. The diver is allowed 5 minutes to descend to 20 fsw, shift to the bottom mixture and perform equipment checks.
6. Have the diver begin descent.
7. Start bottom time.
 - If the diver spends 5 minutes or less performing above procedures, bottom time starts when the diver leaves 20 fsw.
 - If the diver spends more than 5 minutes performing above procedures, bottom time starts at the 5 minute mark.
8. If it is necessary to bring the diver back to the surface from 20 fsw to correct a problem:
 - Shift the diver from the bottom mixture back to air.
 - Ventilate the diver.
 - Confirm the diver is on air.
 - Have the diver begin ascent.
 - When the diver reenters the water, the 5 minute grace period begins again. No adjustment of bottom time is required for the previous exposure at 20 fsw.

12-4.6 Aborting Dive During Descent. Inability to equalize the ears or sinuses may force the dive to be aborted during descent.

1. If it is necessary to bring the diver back to the surface from depths of 100 fsw and shallower:
 - Ensure the diver is in a no-decompression status.
 - If the bottom mixture is 16% oxygen or greater, ascend directly to the surface at 30 fsw/min.
 - If the bottom mixture is less than 16% oxygen, ascend to 20 fsw at 30 fsw/min.
 - Shift the diver from the bottom mixture back to air.
 - Ventilate the diver.
 - Confirm the diver is on air.
 - Complete ascent to the surface on air.
 - If desired, another dive may be performed following a dive aborted 100

fsw and shallower. Add the bottom time of all the dives to the bottom time of the new dive and use the deepest depth when calculating a table and schedule for the new dive.

2. If it is necessary to abort a dive deeper than 100 fsw:
 - Follow the normal decompression schedule to the surface.
 - Repetitive diving is not allowed following a dive aborted deeper than 100 fsw.

12-4.7 Procedures for Shifting to 50 Percent Helium/50 Percent Oxygen at 90 fsw. All dives except no-decompression dives require a shift from bottom mixture to 50% helium 50% oxygen at 90 fsw during decompression. Follow these steps:

1. Shift the console to 50% helium 50% oxygen when the diver reaches 90 fsw.
2. If there is a decompression stop at 90 fsw, ventilate each diver for 20 seconds at 90 fsw.
3. Confirm the divers are on 50% helium 50% oxygen.
4. If there is no decompression stop at 90 fsw, delay ventilation until arrival at the next shallower stop.

Gas shift time is included in the stop time.

12-4.8 Procedures for Shifting to 100 Percent Oxygen at 30 fsw. All in-water decompression dives require a shift to 100 percent oxygen at the 30-fsw stop. Upon arrival at the stop, ventilate each diver with oxygen following these steps:

1. Shift the console to 100% oxygen when the diver reaches 30 fsw.
2. Ventilate each diver for 20 seconds.
3. Verify the diver's voice change.

Time at the 30-fsw stop begins when the divers are confirmed to be on oxygen.

12-4.9 Air Breaks at 30 and 20 fsw. At the 30-fsw and 20-fsw water stops, the diver breathes oxygen for 30-minute periods separated by 5-minute air breaks. The air breaks do not count toward required decompression time. When an air break is required, shift the console to air for 5 minutes then back to 100% oxygen. Ventilation of the divers is not required. For purposes of timing air breaks, begin clocking oxygen time when all divers are confirmed on oxygen. If the total oxygen stop time is 35 minutes or less, an air break is not required at 30 minutes. If the final oxygen period is 35 minutes or less, a final air break at the 30-minute mark is not required. In either case, surface the diver on 100% oxygen upon completion of the oxygen time.

Example

1. Divers are decompressing in the water on a 220 fsw for 20 minute decompression schedule. The schedule calls for 23 minutes on oxygen at 30 fsw and 41 minutes on oxygen at 20 fsw.
2. Divers start their 23 minute 30 fsw stop time when confirmed to be on oxygen at 30 fsw.
3. After 23 minute on oxygen at 30 fsw, divers travel to 20 fsw to complete their 41 minute 20 fsw stop. The 20 second travel time from 30 to 20 fsw on oxygen is included in the 41 minute stop time.
4. Seven minutes from the time the divers left their 30 fsw stop, the console is shifted to air. This is due to the divers having completed a total of 30 minutes on oxygen. No ventilation of the divers is required.
5. After five minutes on air, the console is shifted back to oxygen. No ventilation is required. The five-minute period is considered dead time from the decompression standpoint. A total of 34 minutes on oxygen remain to be completed at 20 fsw.
6. Since the remaining oxygen time is less than 35 minutes, the divers breathe oxygen for the last 34 minutes prior to ascent to the surface without taking an additional air break. Divers remain on oxygen for ascent to the surface.

12-4.10 Ascent from the 20-fsw Water Stop. For normal in-water decompression, the diver surfaces from 20 fsw on oxygen. Ascent rate is 30 fsw/min.

12-4.11 Surface Decompression on Oxygen (SurDO₂). Surface decompression on oxygen is preferred over in-water decompression on oxygen for routine operations. SurDO₂ procedures improve the diver's comfort and safety. A diver is eligible for surface decompression when he has completed the 40-fsw water stop. To initiate surface decompression:

WARNING The interval from leaving 40-fsw in the water to arriving at 50-fsw in the chamber cannot exceed 5 minutes without incurring a penalty. (See paragraph 12-5.14)

1. Bring the diver to the surface at 40 fsw/min and undress him.
2. Place the diver in the recompression chamber. Use of an inside tender when two divers undergo surface decompression is at the discretion of the Dive Supervisor. If an inside tender is not used, both divers will carefully monitor each other in addition to being closely observed by topside personnel.
3. Compress the diver on air to 50 fsw at a maximum compression rate of 100 fsw/min. The surface interval is the elapsed time from the time the diver leaves the 40-fsw water stop to the time the diver arrives at 50 fsw in the chamber. A normal surface interval should not exceed 5 minutes.
4. Upon arrival at 50 fsw, place the diver on 100 percent oxygen by mask. The mask will be strapped on both divers to ensure a good oxygen seal.
5. In the chamber, have the divers breathe oxygen for 30-minute periods separated by 5-minute air breaks. The number of oxygen periods required is indicated

in [Table 12-3](#). The first period consists of 15 minutes on oxygen at 50 fsw followed by 15 minutes on oxygen at 40 fsw. Periods 2, 3, and 4 are spent at 40 fsw. Periods 5, 6, 7 and 8 are spent at 30 fsw. Ascent from 50 to 40 and from 40 to 30 fsw is at 30 fsw/min. Ascent time is included in the oxygen/air time. Ascent from 40 to 30 fsw, if required, should take place during the air break.

6. When the last oxygen breathing period has been completed, return the diver to breathing chamber air.
7. Ascend to the surface at a rate of 30 fsw/min.

The diving supervisor can initiate surface decompression at any point during in-water oxygen decompression at 30 or 20 fsw, if desired. Surface decompression may become desirable if sea conditions are deteriorating, the diver feels ill, or some other contingency arises. Once in the chamber, the diver should receive the full number of chamber oxygen periods prescribed by the tables. Unlike in air diving, no credit is allowed for time already spent on oxygen in the water.

12-4.12 Variation in Rate of Ascent. The rate of ascent to the first stop and between subsequent stops is 30 fsw/minute. Minor variations in the rate of travel between 20 and 40 fsw/minute are acceptable.

12-4.12.1 Early Arrival at the First Stop. If the divers arrive early at the first stop:

1. Begin timing the first stop when the required travel time has been completed.
2. If the first stop requires a gas shift, initiate the gas shift and ventilation upon arrival at the stop, but begin the stop time only when the required travel time has been completed.

12-4.12.2 Delays in Arriving at the First Stop.

1. Delay less than 1 minute. Delays in arrival at the first stop of less than 1 minute may be ignored.
2. Delay greater than 1 minute. Round up the delay time to the next whole minute and add it to the bottom time. Recompute the decompression schedule. If no change in schedule is required, continue on the planned decompression. If a change in schedule is required and the new schedule calls for a decompression stop deeper than the diver's current depth, perform any missed deeper stops at the diver's current depth. Do not go deeper.

Example: If the delay time to arrival at the first stop is 3 minutes and 25 seconds, round up to the next whole minute and add 4 minutes to the bottom time. Recheck the decompression table to see if the decompression stop depths or times have changed.

12-4.12.3 Delays in Leaving a Stop or Arrival at the Next Stop.

■ Delays Deeper than 90 fsw.

1. Delays less than 1 minute may be ignored.

2. Greater than 1 minute. Add the delay to the bottom time and recalculate the required decompression. If a new schedule is required, pick up the new schedule at the present stop or subsequent stop if the delay occurs between stops. Ignore any missed stops or time deeper than the depth at which the delay occurred. If a delay occurs between stops, restart subsequent stop time at completion of the delay.

■ Delays 90 fsw and shallower:

1. Delays less than 1 minute may be ignored.
2. Delays greater than 1 minute require no special action except as described below under special considerations when decompressing with high oxygen partial pressure. Resume the normal decompression schedule at the completion of the delay. If a delay occurs between stops, restart subsequent stop time at completion of the delay.

■ Special considerations when decompressing with high oxygen partial pressure:

1. Delays greater than 5 minutes between 90 and 70 fsw. Shift the diver to air to avoid the risk of CNS oxygen toxicity. At the completion of the delay, return the diver to 50% helium 50% oxygen. Add the time on air to the bottom time and recalculate the required decompression. If a new schedule is required, pick up the new schedule at the present stop or subsequent stop if delay occurs between stops. Ignore any missed stops or time deeper than the depth at which the delay occurred.
2. Delays leaving the 30-fsw stop. Delays greater than 1 minute leaving the 30-fsw stop shall be subtracted from the 20-fsw stop time.

12-4.12.4 **Delays in Travel from 40 fsw to the Surface for Surface Decompression.** Disregard any delays in travel from 40 fsw to the surface during surface decompression unless the diver exceeds the 5-minute surface interval. If the diver exceeds the 5-minute surface interval, follow the guidance in [paragraph 12-5.14](#).

12-5 SURFACE-SUPPLIED HELIUM-OXYGEN EMERGENCY PROCEDURES

In surface-supplied mixed gas diving, specific procedures are used in emergency situations. The following paragraphs detail these procedures. Other medical/physiological factors that surface-supplied mixed gas divers need to consider are covered in detail in [Chapter 3](#). The U.S. Navy Treatment Tables are presented in [Chapter 17](#).

12-5.1 Bottom Time in Excess of the Table.

In the rare instance of diver entrapment or umbilical fouling, bottom times may exceed 120 minutes, the longest value shown in the table. When it is foreseen that bottom time will exceed 120 minutes, immediately contact the Navy Experimental Diving Unit for advice on which decompression procedure to follow. If advice cannot be obtained in time:

1. Decompress the diver using the 120-minute schedule for the deepest depth attained.
2. Shift to 100 percent oxygen at 40 fsw.
3. Surface the diver after completing 30 minutes on oxygen at 40 fsw. Oxygen time at 40 fsw starts when divers are confirmed on oxygen.
4. Compress the diver to 60 fsw in the chamber as fast as possible not to exceed 100 fsw/min.
5. Treat the diver on an extended [Treatment Table 6](#). Extend [Treatment Table 6](#) for two oxygen breathing periods at 60 fsw (20 minutes on oxygen, then 5 minutes on air, then 20 minutes on oxygen) and two oxygen breathing periods at 30 fsw (60 minutes on oxygen, then 15 minutes on air, then 60 minutes on oxygen).

12-5.2 Loss of Helium-Oxygen Supply on the Bottom. Follow this procedure if the umbilical helium-oxygen supply is lost on the bottom:

1. Shift the diver to the emergency gas system (EGS).
2. Abort the dive.
3. Remain on the EGS until arrival at 90 fsw.
4. At 90 fsw, shift the diver to 50% helium 50% oxygen and complete the decompression as planned.
5. If the EGS becomes exhausted before 90 fsw is reached, shift the diver to air, complete decompression to 90 fsw, shift the diver to 50% helium 50% oxygen, and continue the decompression as planned.

12-5.3 Loss of 50 Percent Oxygen Supply During In-Water Decompression. If the diver cannot be shifted to 50% helium 50% oxygen at 90 fsw or the 50% helium 50% oxygen supply is lost during decompression:

1. Shift the diver to air and continue the decompression as planned while trying to correct the problem.
2. Shift the diver to 50% helium 50% oxygen once the problem is corrected. Time spent on air counts toward decompression.
3. If the problem cannot be corrected:
 - Continue the planned decompression on air.
 - Shift the diver from air to oxygen upon arrival at the 50-fsw stop.
 - Breathe oxygen at 50 and 40 fsw for the decompression times indicated in [Table 12-4](#), but not to exceed 16 minutes at 50 fsw. Oxygen time at 50 fsw starts when divers are confirmed on oxygen. If the 50-fsw stop exceeds 16 minutes, travel divers to 40 fsw and add remaining 50-fsw stop time to the 40-fsw stop time on oxygen.

- Surface decompress per [paragraph 12-4.11](#) following completion of the 40-fsw stop.

12-5.4 Loss of Oxygen Supply During In-Water Decompression. If the diver cannot be shifted to oxygen at 30 fsw or the oxygen supply is lost during the 30- or 20-fsw water stops:

1. Switch back to 50% helium 50% oxygen. If a switch to 50% helium 50% oxygen is not possible, switch the diver to air.
2. If the problem can be quickly remedied, reventilate the diver with oxygen and resume the schedule at the point of interruption. Consider any time on helium-oxygen or air as dead time.
3. If the problem cannot be remedied, initiate surface decompression. Ignore any time already spent on oxygen at 30 or 20 fsw. The five minute surface interval requirement for surface decompression begins upon leaving the 30- or 20-fsw stop.
4. If the problem cannot be remedied and surface decompression is not feasible, complete the decompression on 50% helium 50% oxygen or air. For 50% helium 50% oxygen, double the remaining oxygen time at each water stop. For air, triple the remaining oxygen time.

Example: A diver loses oxygen 15 minutes into the 30-fsw water stop and is switched back to the 50% helium 50% oxygen decompression mixture. The problem cannot be corrected. The divers original schedule called for 32 minutes of oxygen at 30 fsw and 58 minutes of oxygen at 20 fsw.

Seventeen minutes of oxygen time ($32 - 15$) remain at 30 fsw. Fifty-eight minutes remain at 20 fsw. The diver should spend an additional 34 minutes (17×2) at 30 fsw on the 50/50 mixture, followed by 116 minutes (58×2) at 20 fsw. Surface the diver upon completion of the 20-fsw stop.

Example: A diver loses oxygen 10 minutes into the 30-fsw water stop and is switched to air. The problem cannot be corrected. The diver's original schedule called for 28 minutes of oxygen at 30 fsw and 50 minutes of oxygen at 20 fsw.

Eighteen minutes of oxygen time ($28 - 10$) remain at 30 fsw. Fifty minutes remain at 20 fsw. The diver should spend an additional 54 minutes (18×3) at 30 fsw on air followed by 150 minutes (50×3) on air at 20 fsw. Surface the diver upon completion of the 20-fsw stop.

12-5.5 Loss of Oxygen Supply in the Chamber During Surface Decompression. If the oxygen supply in the chamber is lost during surface decompression, have the diver breathe chamber air.

- Temporary Loss. Return the diver to oxygen breathing. Consider any time on air as dead time.

- **Permanent Loss.** Multiply the remaining oxygen time by three to obtain the equivalent chamber decompression time on air. If 50% helium 50% oxygen is available, multiply the remaining oxygen time by two to obtain the equivalent chamber decompression time on 50/50. If the loss occurred at 50 or 40 fsw, allocate 10% of the equivalent air or helium-oxygen time to the 40-fsw stop, 20% to the 30-fsw stop, and 70% to the 20-fsw stop. If the diver is at 50 fsw, ascend to 40 fsw to begin the stop time. If the loss occurred at 30 fsw, allocate 30% of the equivalent air or helium-oxygen time to the 30-fsw stop and 70% to the 20-fsw stop. Round the stop times to the nearest whole minute. Surface upon completion of the 20-fsw stop.

Example: The oxygen supply to the chamber is lost 10 minutes into the first 30-minute period on oxygen. Helium-oxygen is not available. The original surface decompression schedule called for three 30-min oxygen breathing periods (total of 90 minutes of oxygen). The diver is at 50 fsw.

The remaining oxygen time is 80 minutes (90-10). The equivalent chamber decompression time on air is 240 minutes (3 x 80). The 240 minutes of air stop time should be allocated as follows: Twenty-four minutes at 40 fsw (240 x 0.1), 48 minutes at 30 fsw (240 x 0.2), and 168 minutes at 20 fsw (240 x 0.7). As addressed above, the diver should ascend from 50 to 40 fsw and begin the 24 minute stop time at 40 fsw.

12-5.6 Decompression Gas Supply Contamination. If the decompression gas supply becomes contaminated with the bottom mixture, 50/50 mix, air, or oxygen:

1. Find the source of the contamination and correct the problem. Probable sources include:
 - An improper valve line-up on the console. This can be verified by checking oxygen percentage on console oxygen analyzer.
 - Accidental opening of the emergency gas supply (EGS) valve.
2. When the problem is corrected:
 - Ventilate each diver for 20 seconds and confirm divers are on decompression gas.
 - Continue decompression as planned. Do not lengthen stop times to compensate for the time spent correcting the problem.

12-5.7 CNS Oxygen Toxicity Symptoms (Nonconvulsive) at the 90-60 fsw Water Stops.

CNS oxygen toxicity symptoms are unlikely but possible while the diver is breathing 50% helium 50% oxygen in the water at depths 60 fsw and deeper. If symptoms of oxygen toxicity do appear, take the following actions:

1. Bring the divers up 10 feet and shift to air to reduce the partial pressure of oxygen. Shift the console as the divers are traveling.

2. Ventilate both divers upon arrival at the shallower stop. Ventilate the stricken diver first.
3. Remain at the shallower stop until the missed time at the previous stop is made up.
4. Resume the planned decompression breathing air.
5. Upon arrival at the next shallower stop, return the divers to the 50% helium 50% oxygen mixture. Ignore any missed time on the 50/50 mixture. A recurrence of symptoms is highly unlikely because of the reduced oxygen partial pressure at the shallower depth.

Example: Red Diver has an oxygen toxicity symptom 5 minutes into his scheduled 9-minute 80-fsw stop. The stage with both divers travels to 70 fsw and the console is shifted to air. Upon arrival at 70 fsw, Red diver is ventilated for 20 seconds followed by Green diver. The divers remain at 70 fsw for the remaining 4 minutes left from their 80-fsw stop and then start their 10 minute scheduled 70-fsw stop time at the completion of the 4 minutes. Upon reaching 60 fsw, the console is shifted back to their 50/50 mixture and both divers are ventilated. The normal decompression schedule is resumed at 60 fsw.

12-5.8 Oxygen Convulsion at the 90-60 fsw Water Stop. If symptoms of oxygen toxicity progress to an oxygen convulsion at 90-60 fsw despite the measures taken above, a serious emergency has developed. Only general management guidelines can be presented here. Topside supervisory personnel must take whatever action they deem necessary to bring the casualty under control.

Follow these procedures when the diver is convulsing at the 90-60 fsw water stops:

1. Shift both divers to air if this action has not already been taken.
2. Have the unaffected diver ventilate himself and then ventilate the stricken diver.
3. If only one diver is in the water, launch the standby diver immediately and have him ventilate the stricken diver.
4. Hold the divers at depth until the tonic-clonic phase of the convulsion has subsided. The tonic-clonic phase of a convulsion generally lasts 1 to 2 minutes.
5. At the end of the tonic-clonic phase, have the dive partner or standby diver ascertain whether the diver is breathing. The presence or absence of breath sounds will also be audible over the intercom.
6. If the diver appears not to be breathing, have the dive partner or standby diver attempt to reposition the head to open the airway. Airway obstruction will be the most common reason why an unconscious diver fails to breathe.
7. If the affected diver is breathing, have the dive partner or standby diver tend the stricken diver and decompress both divers on air following the original schedule. Shift the divers to 50% helium 50% oxygen upon arrival at 50 fsw. Surface decompress upon completion of the 40-fsw water stop.

8. If it is not possible to verify that the affected diver is breathing, leave the unaffected diver at the stop to complete decompression, and surface the affected diver and the standby diver at 30 fsw/min. Shift the unaffected diver back to his 50/50 mixture for completion of decompression. The standby diver should maintain an open airway on the stricken diver during ascent. On the surface the affected diver should receive any necessary airway support and be immediately recompressed and treated for arterial gas embolism and missed decompression in accordance with [Figure 17-1](#).

12-5.9 CNS Toxicity Symptoms (Nonconvulsive) at 50- and 40-fsw Water Stops. It is very unlikely that a diver will develop symptoms of CNS oxygen toxicity while breathing 50% helium 50% oxygen at the 50- and 40-fsw water stops. Symptoms are much more likely if the diver is breathing 100% oxygen in accordance with [paragraph 12-5.3](#). If the diver does experience symptoms of CNS oxygen toxicity at 50 or 40 fsw while breathing either 50% helium 50% oxygen or 100% oxygen, take the following actions:

1. Bring the divers up 10 feet and shift to air to reduce the partial pressure of oxygen. Shift the console as the divers are traveling to the shallower stop.
2. Ventilate both divers upon arrival at the shallower stop. Ventilate the stricken diver first.
3. Remain on air at the shallower depth for double the missed time from 50- and 40-fsw water stops, then surface decompress the diver in accordance with [paragraph 12-4.11](#). If the diver was on 100% oxygen in accordance with [paragraph 12-5.3](#), triple the missed time from the 50- and 40-fsw water stops, then surface decompress.

Example: A diver on 50% helium 50% oxygen experiences an oxygen symptom five minutes into his 10 min stop at 50 fsw. He immediately ascends to 40 fsw and begins breathing air. The decompression schedule calls for a 10 min stop at 40 fsw. The diver missed 5 min of helium-oxygen at 50 fsw and will miss 10 min more at 40 fsw by virtue of the fact that he is on air. The total missed helium-oxygen time is 15 min. The diver should remain at 40 fsw for 30 min, then surface decompress.

Example: A diver on 100% oxygen experiences an oxygen symptom five minutes into his 10 min stop at 40 fsw. He immediately ascends to 30 fsw and begins breathing air. The missed oxygen time at 40 fsw is 5 min. The diver should remain on air at 30 fsw for 15 min, then surface decompress.

4. If surface decompression is not feasible, continue decompression in the water on either air or oxygen depending on the diver's condition:
 - To continue on oxygen, ascend to 30 fsw (or remain at 30 fsw if already there). Take a 10 min period on air (Time on air does not count toward decompression). Then shift the diver to oxygen and complete decompression in the water according to the schedule.

- To continue on air, ascend to 30 fsw (or remain at 30 fsw if already there). Compute the remaining 30- and 20-fsw air stop times by tripling the oxygen time given in the original schedule. Surface upon completion of the 20-fsw stop.
- Alternatively, the diver may complete the 30-fsw stop on air by tripling the oxygen stop time, then switch to oxygen upon arrival at 20 fsw. Remain at 20 fsw for the oxygen time indicated in the original schedule. Surface upon completion of the 20-fsw stop.

12-5.10 Oxygen Convulsion at the 50-40 fsw Water Stop. If oxygen symptoms progress to an oxygen convulsion despite the measures described above or if a convulsion occurs suddenly without warning at 50 or 40 fsw, take the following actions:

1. Shift both divers to air if this action has not already been taken. Have the unaffected diver ventilate himself then ventilate the stricken diver.
2. Follow the guidance given in [paragraph 12-5.8](#) for stabilizing the stricken diver and determining whether he is breathing. If the diver is breathing, hold him at his current depth until he is stable, then take one of the following actions:
 - If the diver missed helium-oxygen or oxygen decompression time at 50 fsw, hold the diver at depth until the total elapsed time on air is at least double the missed time on helium-oxygen, then surface decompress following the steps in [paragraph 12-4.11](#). If the diver was on 100% oxygen in accordance with [paragraph 12-5.3](#), remain at depth until the total elapsed time on air is at least triple the missed time on oxygen, then surface decompress. In either case, add the 40-fsw water stop time to the 50-fsw chamber oxygen stop time.
 - If the diver did not miss any helium-oxygen or oxygen decompression time at 50 fsw, surface decompress following the steps in [paragraph 12-4.11](#). Add any missed oxygen or helium-oxygen time at 40 fsw to the 50-fsw chamber oxygen stop time.
3. If surface decompression is not feasible, complete decompression in the water on air. Compute the remaining stop times on air by doubling the remaining helium-oxygen time, or tripling the remaining oxygen time at each stop.
4. If the diver is not breathing, surface the diver at 30 fsw/min while maintaining an open airway. Treat the diver for arterial gas embolism ([Figure 17-1](#)).

12-5.11 CNS Oxygen Toxicity Symptoms (Nonconvulsive) at 30- and 20-fsw Water Stops. If the diver develops symptoms of CNS toxicity at the 30- or 20-fsw water stops, take the following action:

1. If a recompression chamber is available on the dive station, initiate surface decompression. Shift the console to air during travel to the surface. Once in the chamber, take the full number of chamber oxygen periods prescribed by the tables. Unlike in air diving, no credit is allowed for time already spent on oxygen in the water.

2. If a recompression chamber is not available on the dive station and the event occurs at 30 fsw, bring the divers up 10 fsw and shift to air to reduce the partial pressure of oxygen. Shift the console as the divers are traveling to 20 fsw. Ventilate both divers with air upon arrival at 20 fsw. Ventilate the affected diver first. Complete the decompression on air in the water at 20 fsw. Compute the required air time at 20 fsw by tripling the sum of the missed oxygen time at 30 and 20 fsw.
3. If a recompression chamber is not available on the dive station and the event occurs at 20 fsw, shift the console to air, ventilate both divers, affected diver first, and complete the decompression in the water at 20 fsw on air. Compute the required air time at 20 fsw by tripling the missed oxygen time at 20 fsw.

12-5.12 Oxygen Convulsion at the 30- and 20-fsw Water Stop. If symptoms progress to an oxygen convulsion despite the above measures, a serious emergency has developed and the following actions must be taken.

1. Shift both divers to air and follow the guidance given in [paragraph 12-5.8](#) for stabilizing the diver and determining whether he is breathing.
2. If the diver is breathing, hold him at depth until he is stable, then surface decompress.
3. If surface decompression is not feasible, ventilate both divers with air and complete decompression in the water on air. Compute the remaining stop times on air by tripling the remaining oxygen time at each stop. See [paragraph 12-5.4](#) for example.
4. If the diver is not breathing, surface the diver at 30 fsw/min while maintaining an open airway and treat the diver for arterial gas embolism ([Figure 18-1](#)).

12-5.13 Oxygen Toxicity Symptoms in the Chamber. At the first sign of CNS oxygen toxicity, the patient should be removed from oxygen and allowed to breathe chamber air. Fifteen minutes after all symptoms have completely subsided, resume oxygen breathing at the point of interruption. If symptoms of CNS oxygen toxicity develop again or if the first symptom is a convulsion, take the following action:

1. Remove the mask.
2. After all symptoms have completely subsided, decompress 10 feet at a rate of 1 fsw/min. For a convulsion, begin travel when the patient is fully relaxed and breathing normally.
3. Resume oxygen breathing at the shallower depth at the point of interruption.
4. If another oxygen symptom occurs, complete decompression on chamber air. Follow the guidance given in [paragraph 12-5.5](#) for permanent loss of chamber oxygen supply to compute the air decompression schedule.

12-5.14 Surface Interval Greater than 5 Minutes. If the time from leaving 40 fsw in the water to the time of arrival at 50 fsw in the chamber during surface decompression exceeds 5 minutes, take the following action:

1. If the surface interval is less than or equal to 7 minutes, add one-half oxygen period to the total number of chamber periods required by increasing the time on oxygen at 50 fsw from 15 to 30 minutes. Ascend to 40 fsw during the subsequent air break. The 15-min penalty is considered a part of the normal surface decompression procedure, not an emergency procedure.
2. If the surface interval is greater than 7 minutes, continue compression to a depth of 60 fsw. Treat the divers on [Treatment Table 5](#) if the original schedule required 2 or fewer oxygen periods in the chamber. Treat the divers on [Treatment Table 6](#) if the original schedule required 3 or more oxygen periods in the chamber.
3. On rare occasions a diver may be unable to reach 50 fsw in the chamber due to difficulty equalizing middle ear pressure. In this situation, an alternative procedure for surface decompression on oxygen may be used:
 - Begin oxygen breathing at the initially attained depth - presumed to be less than 20 fsw.
 - If surface decompression was initiated while the diver was decompressing on oxygen in the water at 20 fsw, attempt to gradually compress the diver to 20 fsw.
 - If surface decompression was initiated from deeper than 20 fsw, attempt to gradually compress the diver to 30 fsw.
 - In either case, double the number of chamber oxygen periods indicated in the table and take these periods at the deepest depth the diver is able to attain. Oxygen time starts when the diver initially goes on oxygen.
 - Interrupt oxygen breathing every 60 minutes with a 15-min air break. The air break does not count toward the total oxygen time.
 - Surface the diver at 30 fsw/min upon completion of the oxygen breathing periods and carefully observe the diver for the onset of decompression sickness.

This “safe way out” procedure is not intended to be used in place of normal surface decompression procedures. Divers that experienced ear difficulty on descent in the water column may not be good candidates for surface decompression.

- 12-5.15 Asymptomatic Omitted Decompression.** Certain emergencies may interrupt or prevent required decompression. Unexpected surfacing, exhausted gas supply and bodily injury are examples of such emergencies. [Table 12-3](#) shows the initial management steps to be taken when the diver has an uncontrolled ascent.

Table 12-3. Management of Asymptomatic Omitted Decompression.

Deepest Decompression Stop Omitted	Decompression Status	Surface Interval (Note 1)	ACTION
None	No-D	Any	Observe on surface for one hour
20 or 30 fsw	Stops Required	Less than 1 min	Return to depth of stop. Increase stop time by 1 min. Resume decompression according to original schedule
		1–7 min	Use Surface Decompression Procedure (Note 2)
		Greater than 7 min	Treatment Table 5 if 2 or fewer SurDO ₂ periods Treatment Table 6 if 3 or more SurDO ₂ periods
40 or 50 fsw	Stops Required	Any	Treatment Table 6
Deeper than 50 fsw	Stops Required: Less than 60 min missed	Any	Treatment Table 6A
	Stops Required: More than 60 min missed	Any	Compress to depth of dive not to exceed 225 fsw. Use Treatment Table 8 For saturation systems: Compress to depth of dive. Saturate two hours. Use saturation decompression without an initial upward excursion
Notes: 1. Surface interval is the time from leaving the stop to arriving at depth in the chamber. 2. Using a recompression chamber is strongly preferred over in-water recompression for returning a diver to pressure. Compress to depth as fast as possible not to exceed 100 fsw/min. 3. For surface intervals greater than 5 minutes but less than or equal to 7 minutes, increase the oxygen time at 50 fsw from 15 to 30 minutes.			

- 12-5.15.1 Omitted Decompression Stop Deeper Than 50 fsw.** An omitted decompression stop deeper than 50 fsw when more than 60 minutes of decompression are missed is an extreme emergency. The diver shall be returned as rapidly as possible to the full depth of the dive or the deepest depth of which the chamber is capable, whichever is shallower.

- 12-5.15.1.1 For Nonsaturation Systems.** For nonsaturation systems, the diver shall be rapidly compressed on air to the depth of the dive or to 225 feet, whichever is shallower.